Capstone Project

Fuel Finder App

Project Documentation

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1. Introduction

In Ethiopia, the traditional way of acquiring fuel is often challenging and inefficient. When drivers need to refuel their vehicles, they typically have to physically drive from one station to another, without any prior information about fuel availability. During the frequent fuel shortages that occur throughout the country, this process becomes even more difficult. Drivers might visit multiple stations only to find empty pumps or long queues that can stretch for hours. Information about which stations have fuel is usually spread by word of mouth, with people calling friends or family to ask about fuel availability in different areas. This unreliable system leads to significant waste of time, money, and the very fuel drivers are trying to conserve. Some people resort to storing extra fuel at home in unsafe containers, while others plan their entire day around the possibility of finding fuel. Business operations and daily commutes are often disrupted by these unpredictable shortages.

The Fuel Finder App addresses these challenges by providing a comprehensive solution that leverages technology to transform how drivers in Ethiopia locate available fuel. The app uses location services to help users find nearby operational fuel stations with real-time information about fuel availability. Available as both a Web App and Mobile App, users can quickly check which stations have fuel before starting their journey, saving them valuable time and resources. The app aims to improve the overall fuel purchasing experience across Ethiopia, reducing frustration and increasing efficiency through technological innovation that puts reliable information directly in the hands of drivers.

2. Business Need and Opportunity

2.1. Client Needs

1. Accurate Information

In Ethiopia, drivers frequently face the frustration of not knowing which fuel stations have supply. Without reliable information, they often arrive at stations only to find empty pumps. The Fuel Finder App addresses this by providing verified, real-time updates about fuel availability at different stations. This accurate information allows drivers to make smart decisions about where and when to refuel, eliminating guesswork and disappointment.

2. Time Efficiency

Ethiopians currently waste significant time driving from one empty station to another during fuel shortages. This lost time affects both personal schedules and business operations. The app dramatically improves efficiency by showing users exactly which stations have fuel before they begin their journey. Instead of spending hours searching, drivers can go directly to stations with available supply, reclaiming valuable time for work, family, or other activities.

3. Fuel Conservation

The current practice of searching for fuel ironically wastes the very resource drivers are seeking. Many people drive long distances checking multiple stations, burning through their remaining fuel in the process. The Fuel Finder App helps users conserve fuel by directing them only to stations with available supply. This reduces unnecessary travel, lowers expenses, and decreases the environmental impact of excessive driving.

4. Geographical Coverage

Fuel shortages affect all regions of Ethiopia, from major cities to remote rural areas. A useful solution must work throughout the entire country, not just in urban centers. The Fuel Finder App is designed to cover all areas where there are fuel stations, ensuring that drivers in less populated regions benefit from the same information as those in cities. This comprehensive coverage helps address fuel access inequalities across different geographical areas.

5. Easy Station Management

For the system to work effectively, fuel station operators need a simple way to update their supply status. Many station managers have limited technical expertise and busy schedules. The app provides these operators with an intuitive interface to quickly update fuel availability

information with minimal training or time investment. This simplicity encourages consistent updating, which is essential for maintaining the system's reliability.

6. Cross-Platform Accessibility

Ethiopians access the internet through various devices, including basic mobile phones, smartphones, and computers. To be truly useful, the fuel finder service must be accessible across all these platforms. The solution is available as both a mobile app for smartphone users and a web application for those using computers or basic phones with internet access. This flexibility ensures that all drivers can access the service regardless of their technology resources or location.

2.2. Problems

1. Frequent Fuel Shortages

Ethiopia faces persistent fuel supply challenges throughout the year. Unlike occasional shortages experienced in some countries, Ethiopian drivers deal with recurring periods when fuel is scarce across entire regions. These shortages can last for days or even weeks, creating significant disruption to daily life. During these periods, only a small percentage of stations receive fuel deliveries, but there's no efficient way for drivers to know which ones have supply. This unpredictability forces drivers to develop frustrating coping strategies like constantly keeping their tanks near full or maintaining networks of contacts to share information about fuel availability.

2. Lack of Information Systems

Currently, there is no centralized or reliable method for Ethiopians to check fuel availability before driving to a station. Information typically spreads through informal channels like phone calls to friends, social media posts, or word of mouth. This fragmented approach creates information gaps and often leads to the spread of outdated or incorrect details. Without a trusted system to verify and distribute accurate information, drivers must rely on luck or spend considerable effort gathering intelligence about which stations might have fuel on any given day.

3. Wasted Resources

The traditional approach to finding fuel during shortages involves physically driving from station to station until finding one with available supply. This method wastes considerable resources - the very fuel drivers are trying to conserve, money spent on unnecessary travel, and valuable time that could be used productively elsewhere. For taxi drivers and delivery services, this problem directly impacts their livelihoods. For regular commuters, it creates added stress and

financial burden. The cumulative economic impact of these wasted resources across millions of drivers is substantial.

4. Manual Reporting Challenges

Existing attempts to share fuel availability information rely on manual, inconsistent reporting methods. Some communities use WhatsApp groups or social media, but these updates are often unreliable, quickly outdated, or limited to small geographic areas. Station owners have no standardized way to announce when they receive fuel or when they run out. Without verification mechanisms, false information can spread quickly, sending drivers on fruitless journeys. The lack of incentives for accurate reporting further complicates these manual systems.

5. Digital Divide

Ethiopia has varying levels of technology adoption and internet connectivity across different regions. While urban areas have increasingly reliable internet access and smartphone usage, many rural areas face connectivity challenges and lower rates of digital literacy. Any solution to the fuel information problem must address this digital divide to avoid creating a system that only benefits technologically advantaged populations. The disparity in access to digital tools could potentially worsen fuel access inequality if not properly addressed.

6. Transportation Inefficiency

Beyond personal inconvenience, fuel shortages and the inability to quickly locate available fuel significantly impact essential services and business operations. Emergency vehicles, public transportation, delivery services, and businesses that rely on regular transportation face operational disruptions when fuel cannot be located efficiently. These inefficiencies ripple through the economy, affecting productivity, increasing costs, and sometimes compromising critical services. During severe shortages, entire supply chains can be disrupted, impacting food delivery, healthcare access, and other essential services.

3. Objective

3.1. General Objective

To develop and deploy a comprehensive fuel finder application that helps drivers in Ethiopia locate available nearby fuel stations in real-time.

3.2. Specific Objectives

1. Design and implement a user-friendly web and mobile application to locate nearby fuel stations.
2. Develop a real-time fuel availability tracking system of fuel stations with manual updates from station owners.
3. Create a scalable cloud infrastructure to support application deployment and growth.
4. Implement secure user authentication and data management systems.
5. Design an intuitive interface for fuel station owners to update availability information.
6. Design an intuitive interface for Ministry of Mines and Petroleum delegate to set a price and monitor gas station
7. Establish a foundation for future integration with automated fuel level sensors.

4. Constraints and Assumptions

4.1. Business and Technical Constraints

1. **Time Constraint**: The project must adhere to a strict timeline of six weeks, from initial documentation to deployment of the MVP (Minimum Viable Product) version. Timely milestones are essential to meet stakeholder expectations.
2. **Budget Limitations**: Financial resources available for development are constrained, costs may arise for the integration of IoT systems, such as ultrasonic sensors to monitor fuel tanker levels. Furthermore, external service providers, like SMS OTP verification systems and other third-party services, might require fees, which need to be carefully considered in the budget.
3. **Technical Expertise**: The project team has limited experience in operating advanced IoT systems, which may affect the implementation of certain technological aspects.
4. **Initial Manual Updates**: The MVP version relies heavily on manual data input by fuel station owners, as automated updates and synchronization will not be fully operational at launch.
5. **Regulatory Compliance**: The project cannot meet requirements for gathering and licensing regulatory compliance within the given six-week timeframe. Obtaining the necessary licenses and ensuring full adherence to data privacy and security regulations requires additional time for legal reviews, stakeholder approvals, and implementation of compliant features, which is beyond the current schedule.
6. **Language and Localization**: The application will support only two languages, as the integration of professional translations requires specialized expertise and resources.

7. **Device Accessibility**: The app must account for varying levels of smartphone usage and mobile data access among its target users, addressing potential barriers to adoption.
8. **Third-Party Dependency Constraints**: The project relies on external services like Google Maps or OpenStreetMap, which could face issues or policy changes, potentially affecting app functionality.
9. **Hardware Constraints**: Dependency on specific hardware, such as ultrasonic sensors, introduces risks related to availability, compatibility, or costs.
10. **Environmental Constraints**: External factors like power outages, unreliable internet in certain regions, or severe weather conditions could impact app deployment and user experience.
11. **Stakeholder Constraints**: Resistance or lack of active support from stakeholders, including fuel station owners and regulatory authorities, could hinder the project's success.

4.2. Assumptions

1. **Fuel Station Owners' Participation**: It is assumed that fuel station owners will actively engage with the platform and regularly update their fuel availability status. Their consistent input is critical for ensuring accurate, real-time information for users.
2. **User Device Access**: Users are expected to have access to smartphones or internet-enabled devices, as these are essential for interacting with the application. This assumption is crucial for the platform's adoption and usability.
3. **Mapping Service Availability**: It is assumed that Google Maps or OpenStreetMap will provide adequate and reliable coverage of Ethiopian roads and locations. The accuracy and comprehensiveness of these mapping services are fundamental for location-based functionalities.
4. **Internet Connectivity**: There is an assumption that sufficient internet connectivity exists in most urban and semi-urban areas where the app will primarily operate. The availability of stable internet connections is necessary for real-time updates and user interactions.
5. **User Location Sharing**: Users are presumed to be willing to share their location data to receive personalized information about nearby fuel stations. This feature is central to delivering tailored recommendations.
6. **Governmental Support**: It is assumed that the Ministry of Fuel and Mine, as well as other relevant governmental entities, will support the initiative to promote digital innovation in the Fuel and Energy Sector. Such backing is vital for smooth implementation and growth.
7. **Basic Digital Literacy Among Station Owners**: The project assumes that fuel station owners possess basic digital literacy skills, enabling them to interact with the management portal effectively and update information as required.
8. **Six-Week Development Timeline Feasibility**: The six-week timeframe is assumed to be sufficient to develop and deploy the MVP version of the application, considering the scope and resource allocation.

5. Business Scope

5.1. Epics

5.1.1. User Stories

Epic 1: User Registration and Authentication

- As a driver, I want to create an account so that I can personalize my experience and save preferences.
- As a driver, I want to verify my email address during account registration, so that my account is secure and my identity is confirmed.
- As a fuel station owner, I want to register my station so that I can provide accurate, real-time updates.
- As a user, I want to log in securely across multiple devices to access my information seamlessly.
- As an admin, I want register the ministry delegate to verify and monitor the stations

Epic 2: Gas station owner verification

- As an admin, I want to verify fuel station registrations to maintain system accuracy and data quality.

Epic 3: Fuel Station Discovery

- As a driver, I want to see fuel stations on an interactive map so that I can easily locate them based on proximity.
- As a driver, I want to filter fuel stations by fuel type (e.g., diesel, petrol) to find exactly what I need without wasting time.

Epic 4: Fuel Availability Tracking

- As a driver, I want to see real-time fuel availability at stations so that I can avoid wasted trips.
- As a station owner, I want a simple interface to update fuel levels so that customers are always informed.

Epic 5: Rate,Favourite and Feedback

- As a driver, I want to give rating and feedback about inaccurate information to improve service of gas stations.

- As a driver, I want to add gas stations to my favourites list so that I can easily access stations I prefer or frequently use.
- As a station owner, I want to see user feedback given to my station and I respond to feedback manually to improve the service.

Epic 6: view report

- As a ministry delegate, I want to view the stations report

Epic 7: User Privacy

- As a user, I want my location data to be securely handled and used only for fuel station recommendations so that I feel safe using the app.

5.1.2. Acceptance Criteria

Epic 1: User Registration and Authentication

- Users can create accounts using email or phone number
- Account verification via email confirmation
- Secure password recovery process
- Station owners can submit registration documentation

Epic 2: Gas station owner verification

- Admin dashboard for gas station verification

Epic 3: Fuel Station Discovery

- Map displays stations
- Filter options for fuel types (petrol, diesel)

Epic 4: Fuel Availability Tracking

- Real-time availability status indicators (yes/no)
- Simple interface for station owners to update Availability
- Timestamp for last update to show information freshness
- Future support for automated sensor integration

Epic 5: Rate, Favourite and Feedback

- Drivers can submit star ratings and written feedback on gas stations.
- Drivers can add and remove stations from a personalized favourites list.

- Station owners can view user feedback and manually post responses.

Epic 6: View Report

- Ministry delegates can access and view detailed reports of all gas stations, including fuel availability, ratings, and feedback.

Epic 7: User Privacy

- Location data is stored securely using encryption protocols and is not shared with unauthorized third parties.
- The app uses location data exclusively to suggest nearby fuel stations and does not track users beyond this purpose.

6. Technical Scope (Use Case Diagram)

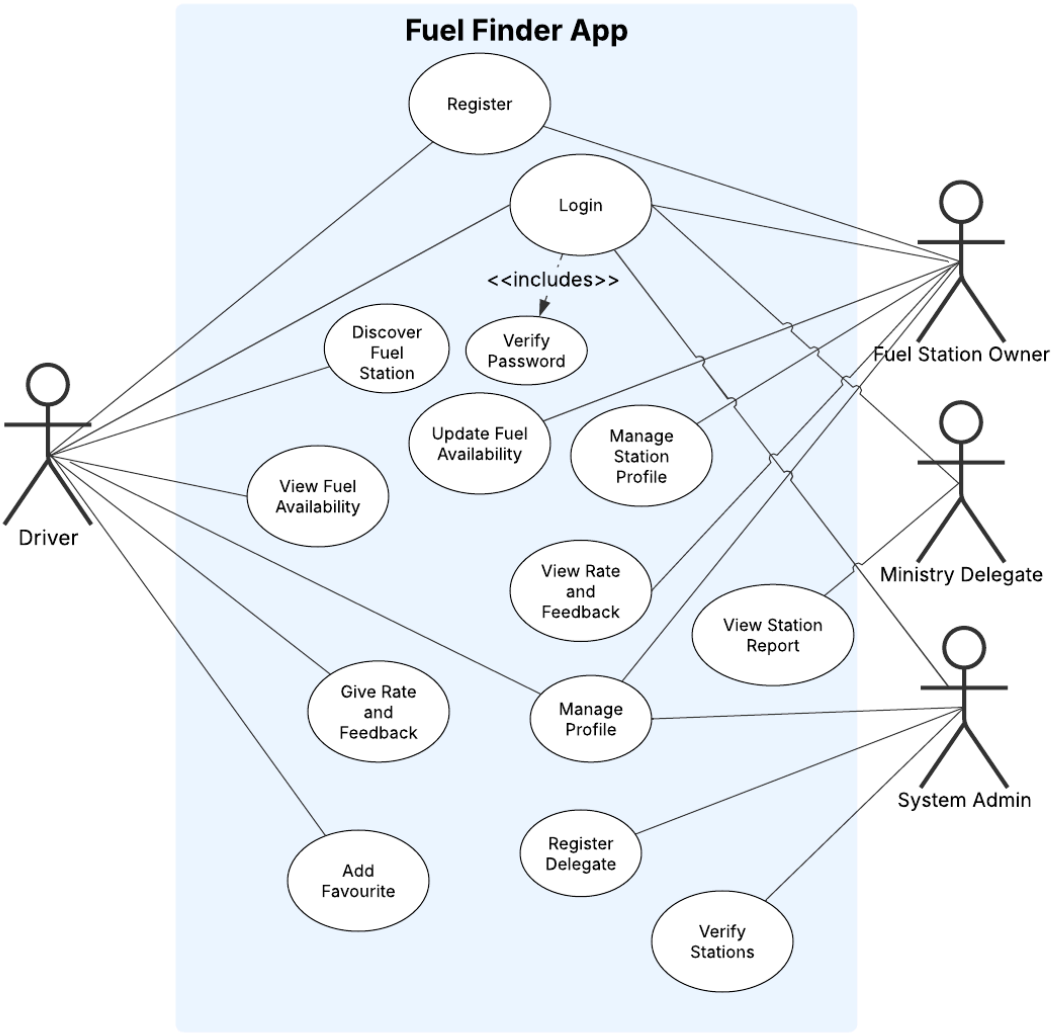
The Use Case Diagram for the Fuel Finder App includes the following actors and use cases:

Actors:

- Driver/User
- Fuel Station Owner
- System Administrator
- Ministry delegate

Primary Use Cases:

1. User Registration and Authentication
2. Locate Nearby Fuel Stations
3. View Fuel Availability
4. Update Fuel Availability (Station Owner)
5. Report Issues
6. Manage Station Profile (Station Owner)
7. Verify Stations (Admin)
8. Generate Reports (Admin)
9. Control and monitoring stations(Ministry delegate)



7. Applied Technology

Frontend:

- React.js for web application
- Flutter for cross-platform mobile application
- Material UI and Native Base for responsive design
- Mapbox ,openStreetMap or Google Maps API for location services

Backend:

- Node.js with Express.js framework
- Postgres for scalable NoSQL database

- JWT for secure authentication

DevOps:

- Docker for containerization
- GitHub Actions for CI/CD pipeline

APIs and Services:

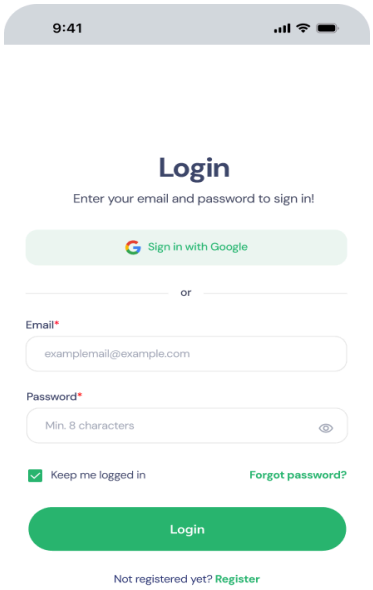
- Geolocation APIs (Google Maps, OpenStreetMap)
- Gemini Api for gas Station suggestion

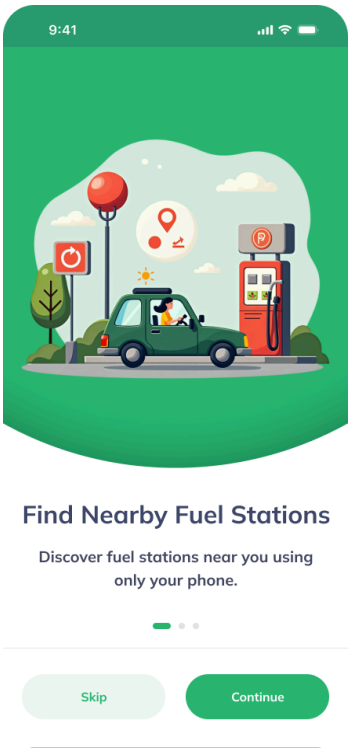
Future Integration:

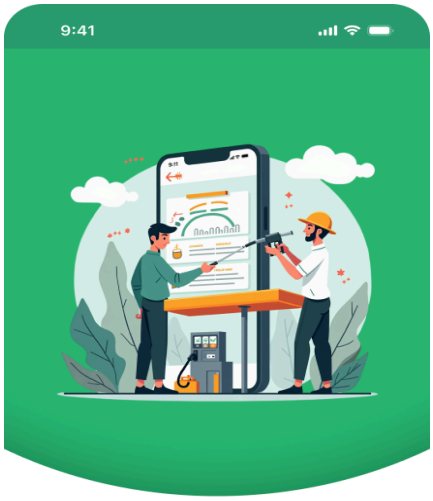
- IoT platform for ultrasonic sensor integration
- MQTT protocol for sensor data transmission

8. User Interface

Mobile app





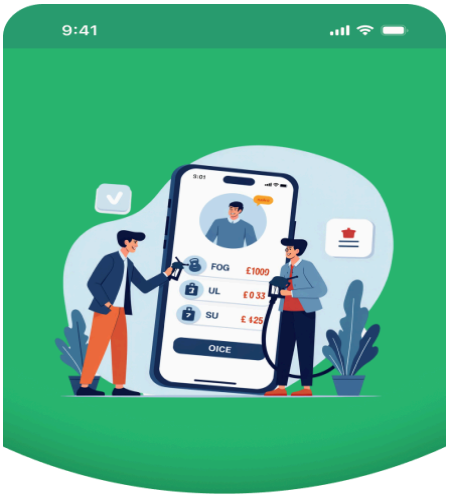


View Fuel Availability

check real-time fuel availability before you go to the station.

Skip

Continue



View Official Fuel Prices

Stay informed with up-to-date fuel prices across all stations.

Get Started

9. Roadmap and Release Plan

Week	Dates	Tasks
Week 1	April 8-14, 2025	- Complete project documentation - Define detailed requirements - Create wireframes and mockups
Week2	April 15-21, 2025	- Design database schema - Implement user authentication system - Develop station registration functionality - Create basic map interface - Set up development environment
Week3	April 22-28, 2025	- Implement fuel availability tracking - Develop station owner portal - Create notification system - Build search and filter functionality - Implement basic analytics
Week4	April 29-May 5, 2025	- Create cross-platform compatibility - Design responsive UI elements - Add push notification functionality
Week5	May 6-13,2025	- Conduct comprehensive testing - Fix identified bugs and issues - Optimize performance
Week6	May 13-19, 2025	- Set up cloud infrastructure - Establish CI/CD pipeline - Deploy to production environment - Conduct final tests